

LITCHFIELD, IL, USA

WMO: 722972

Lat: 39.163N

Lon: 89.675W

Elev: 210

StdP: 98.82

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.9	-12.7	-20.7	0.6	-14.6	-17.8	0.8	-12.4	12.9	-1.3	12.0	-0.1	3.3	310	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.0	23.7	32.5	23.7	31.3	23.3	26.0	31.1	25.2	30.4	24.4	29.4	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.3	29.3	23.7	19.0	28.2	22.8	18.0	27.2	82.0	31.3	78.0	30.5	74.7	29.1	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	9.8	8.6	-19.3	35.5	3.7	2.3	-21.9	37.2	-24.1	38.6	-26.1	39.9	-28.8	41.6		
			-19.8	27.6	3.5	0.8	-22.3	28.1	-24.4	28.6	-26.4	29.1	-28.9	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	-1.5	0.0	6.6	12.5	18.1	23.1	24.3	23.7	20.2	13.3	6.4	0.6
	DBStd	10.53	6.34	6.38	6.24	5.15	4.84	2.92	3.10	3.12	4.07	5.35	5.38	5.32
	HDD10.0	1269	357	284	143	33	3	0	0	0	0	25	131	293
	HDD18.3	2853	614	513	366	181	67	2	1	1	27	173	358	549
	CDD10.0	2128	2	4	37	108	253	393	445	426	305	129	23	2
	CDD18.3	669	0	0	2	7	58	146	187	169	82	18	0	0
	CDH23.3	6100	0	0	11	74	464	1272	1847	1539	739	153	2	0
	CDH26.7	2276	0	0	1	10	126	459	754	613	273	40	0	0
Wind	WSAvg	3.8	4.8	4.5	4.7	4.7	3.9	3.2	2.5	2.3	2.7	3.8	4.4	4.5
Precipitation	PrecAvg	996	67	50	71	101	132	97	84	81	75	80	93	59
	PrecMax	1295	197	113	140	279	291	199	155	145	225	164	185	182
	PrecMin	819	13	12	15	24	32	45	30	37	5	29	13	20
	PrecStd	141	42	30	34	60	80	44	36	30	52	36	46	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	18.5	25.0	27.7	32.3	33.8	37.6	37.0	33.9	30.6	22.6	16.7
		MCWB	13.6	14.1	17.5	17.7	21.7	23.4	23.8	22.6	21.7	21.8	16.1	14.5
	2%	DB	13.0	15.4	22.1	25.8	29.8	32.5	34.0	34.1	32.0	27.4	19.1	13.1
		MCWB	11.1	12.4	16.4	17.6	21.6	23.6	24.5	23.2	22.2	19.3	14.8	10.8
	5%	DB	10.3	12.4	18.8	23.0	27.8	31.2	32.3	32.2	30.1	24.7	17.1	10.7
		MCWB	8.4	9.9	14.7	16.1	20.4	23.2	24.7	23.6	21.8	17.7	13.3	8.9
	10%	DB	7.1	9.0	16.2	21.2	26.1	29.6	30.9	30.1	27.6	22.1	14.7	8.0
		MCWB	5.4	6.8	12.7	15.5	19.7	22.3	24.1	23.1	20.0	16.6	11.5	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.3	18.9	20.3	23.9	26.1	27.2	27.2	25.2	22.8	18.2	14.9
		MCDB	16.0	17.4	23.2	25.9	29.4	31.1	32.2	31.7	30.1	28.6	20.6	16.3
	2%	WB	11.8	13.1	16.8	18.8	22.7	25.0	26.3	25.7	23.8	21.2	15.7	11.8
		MCDB	12.9	15.3	21.3	23.2	27.8	30.5	31.5	30.6	29.1	25.5	18.4	12.7
	5%	WB	8.5	10.1	15.2	17.5	21.7	24.1	25.5	24.8	22.7	19.6	13.8	8.9
		MCDB	9.8	12.3	18.0	21.6	26.4	29.3	30.6	29.9	27.5	22.3	16.3	10.4
	10%	WB	5.2	7.0	13.3	16.0	20.6	23.2	24.7	23.8	21.7	17.7	11.8	6.7
		MCDB	7.2	9.1	15.9	20.1	24.8	28.2	29.5	28.5	26.0	20.8	14.4	8.1
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	9.5	10.9	10.4	10.4	10.6	11.0	12.1	11.4	9.4	7.2
		MCDBR	12.1	13.1	12.6	13.6	12.0	11.9	12.1	12.9	13.9	13.8	12.7	10.7
		MCWBR	10.4	10.3	8.2	7.6	5.6	5.2	5.0	5.1	5.7	6.8	8.8	9.0
	5% WB	MCDBR	11.5	12.4	11.3	11.6	10.6	10.6	10.6	10.9	11.4	11.5	11.1	9.9
		MCWBR	10.5	10.5	7.9	7.9	5.7	5.2	5.2	5.4	5.5	6.5	8.8	9.0
Clear-Sky Solar Irradiance	taub	0.299	0.310	0.344	0.392	0.428	0.428	0.462	0.449	0.405	0.351	0.306	0.295	
	taud	2.469	2.429	2.404	2.269	2.224	2.260	2.157	2.184	2.289	2.449	2.518	2.497	
	Ebn at Noon	878	915	913	884	855	853	819	820	836	855	864	856	
	Edh at Noon	79	94	108	131	140	135	149	141	120	92	75	72	
All-Sky Solar Radiation	RadAvg	1.95	2.74	3.70	4.90	5.52	6.35	6.25	5.58	4.72	3.34	2.24	1.64	
	RadStd	0.22	0.27	0.30	0.29	0.46	0.47	0.42	0.34	0.32	0.36	0.23	0.19	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+82

Nomenclature: See separate page